

Problema 1.2.

(a) Se considera functia compus a $h(t) = g(f(t))$. Sa se exprime conditionarea lui h i functie de conditionarea lui g s,i f. Atentie la formulare - precizati in care puncte se vor evalua numerele de conditionare.

(b) Ilustrati (a) pentru $h(t) = \frac{1+\sin(t)}{1-\sin(t)}$, $t = \pi/4$.

a) Fie functia f :

$$f: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

Conditionarea lui $f = \|\Gamma(x)\|$ unde :

$$\Gamma(x) = \begin{pmatrix} \frac{x_1 \frac{\partial f_1}{\partial x_1}}{f_1(x)} & \dots & \frac{x_m \frac{\partial f_1}{\partial x_m}}{f_1(x)} \\ \vdots & \ddots & \vdots \\ \frac{x_1 \frac{\partial f_n}{\partial x_1}}{f_n(x)} & \dots & \frac{x_m \frac{\partial f_n}{\partial x_m}}{f_n(x)} \end{pmatrix}$$

Fara a reduce generalitatea pentru $m=n=1$ $(\text{cond } f)(x) = \left| \frac{xf'(x)}{f(x)} \right|$.

$$\text{Deci } (\text{cond } h)(t) = \left| \frac{t h'(t)}{h(t)} \right| = \left| \frac{t g'(f(t))}{g(f(t))} \right| = \left| \frac{t * g'(f(t)) * f'(t)}{g(f(t))} \right| = \left| \frac{f(t) * g'(f(t)) * t * f'(t)}{f(t) * g(f(t))} \right| = \left| \frac{f(t) g'(f(t))}{g(f(t))} \right| * \left| \frac{t f'(t)}{f(t)} \right| = (\text{cond } g)(f(t)) * (\text{cond } f)(t).$$

$$\text{b) } (\text{cond } h)(t) = \left| \frac{t * h'(t)}{h(t)} \right|$$

$$h\left(\frac{\pi}{4}\right) = \frac{1 - \sin\left(\frac{\pi}{4}\right)}{1 + \sin\left(\frac{\pi}{4}\right)} = \frac{1 - \frac{\sqrt{2}}{2}}{1 + \frac{\sqrt{2}}{2}} = \frac{1 + 2 * \frac{\sqrt{2}}{2} + 2}{1 - \frac{2}{4}} = \frac{3 + 2\sqrt{2}}{2} * \frac{2}{1} = 3 + 2\sqrt{2}$$

$$h'(t) = \frac{\left(\frac{1+\sin(t)}{1-\sin(t)}\right)'}{\left(\frac{1+\sin(t)}{1-\sin(t)}\right)^2} = \frac{(1+\sin(t))' \cdot (1-\sin(t)) - (1-\sin(t)) \cdot (1+\sin(t))'}{(1-\sin(t))^2} = \frac{\cos(t) - \cos(t) \sin(t) + \cos(t) + \sin(t)}{(1-\sin(t))^2} = \frac{2\cos(t)}{(1-\sin(t))^2}$$

$$h'\left(\frac{\pi}{4}\right) = \frac{\frac{\sqrt{2}}{2}}{1 - \frac{\sqrt{2}}{2} + \frac{2}{4}} = \frac{\frac{\sqrt{2}}{2}}{1 - \frac{\sqrt{2}}{2} + \frac{1}{2}} = \frac{\frac{\sqrt{2}}{2}}{\frac{2 - \sqrt{2} + 1}{2}} = \frac{\sqrt{2}}{3 - 2\sqrt{2}} = \frac{2\sqrt{2}}{3 - 2\sqrt{2}} = \frac{6\sqrt{2} + 8}{9 - 8} = 6\sqrt{2} + 8$$

$$(\text{cond } h)\left(\frac{\pi}{4}\right) = \left| \frac{\frac{\pi}{4} \cdot (6\sqrt{2} + 8)}{3 + 2\sqrt{2}} \right| = \left| \frac{\frac{\pi}{4} \cdot (6\sqrt{2} + 8) \cdot (3 - 2\sqrt{2})}{9 - 8} \right| = \left| \frac{\pi}{4} \cdot (18\sqrt{2} - 24 + 24 - 16\sqrt{2}) \right| = \frac{\pi\sqrt{2}}{2}$$

Se observa functiile : $\mathbb{R} \rightarrow \mathbb{R}, f(t) = \sin(t)$ si $g: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R} \quad g(t) = \frac{1+t}{1-t}$ cu $g(f(t)) = \frac{1+\sin(t)}{1-\sin(t)} = h(t)$.

$$f'\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$(\text{cond } f)\left(\frac{\pi}{4}\right) = \left| \frac{\frac{\pi}{4} \cdot \frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \right| = \frac{\pi}{4}$$

$$g\left(f\left(\frac{\pi}{4}\right)\right) = \frac{1+\sin\left(\frac{\pi}{4}\right)}{1-\sin\left(\frac{\pi}{4}\right)} = \frac{\left(1+\frac{\sqrt{2}}{2}\right)^2}{\left(1-\frac{\sqrt{2}}{2}\right)\left(1+\frac{\sqrt{2}}{2}\right)} = \frac{3+2\sqrt{2}}{\frac{1}{2}} = 3 + 2\sqrt{2}$$

$$g'(t) = \frac{(1+t)' \cdot (1-t) - (1+t)(1-t)'}{(1-t)^2} = \frac{(1-t) + (1+t)}{(1-t)^2} = \frac{2}{(1-t)^2}$$

$$g'\left(f\left(\frac{\pi}{4}\right)\right) = \frac{2}{\left(1-\frac{\sqrt{2}}{2}\right)^2} = \frac{2(3+2\sqrt{2})}{9-8} = 6 + 4\sqrt{2}$$

$$(\text{cond } g)\left(f\left(\frac{\pi}{4}\right)\right) = \left| \frac{\frac{\sqrt{2}}{2} \cdot (6 + 4\sqrt{2})}{3 + 2\sqrt{2}} \right| = \left| \frac{\sqrt{2}(3 + 2\sqrt{2})}{3 + 2\sqrt{2}} \right| = \sqrt{2}$$

$$(\text{cond } g)\left(f\left(\frac{\pi}{4}\right)\right) * (\text{cond } f)\left(\frac{\pi}{4}\right) = \frac{\pi\sqrt{2}}{2} = (\text{cond } h)\left(\frac{\pi}{4}\right)$$